Appendix C

UTILITY THEORY

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Contents

C UTILITY THEORY 1

C.1 Investor Preferences and Utility Functions 4

C.2 Concave Utility Functions and Risk Aversion 6

C.3 Certainty Equivalent Value and the Risk Premium 7

C.4 Absolute and Relative Risk Aversion 8

C.5 Expected Utility Maximization and Mean Variance Optimization 13

C.6 Expected Quadratic Utility and QP Optimization 16

C.7 Objections to Utility Theory 17

C.8 Making Utility Theory Practical 19

C.1 Investor Preferences and Utility Functions

The question of how to choose among investment opportunities (or, more generally, among uncertain prospects) has a rich history, with interdisciplinary contributions by gamblers, economists, mathematicians and psychologists[1]. Early on, it was broadly accepted that the fair price of entry to a game of chance was the expected value of its payoff. By way of example, if the flip of a coin returned \$0 or \$2 with equal probability, the fair price for participation in this game was held to be $0.5 \cdot \$0 + 0.5 \cdot \$2 = \$1$. But in 1710, Nicolaus Bernoulli described a simple game, now known as the St. Petersburg game, that threw this belief into doubt. The St. Petersburg game is played as follows:

A fair coin is repeatedly flipped, and the number of flips till the first appearance of a head, which we denote by k, is noted. The gambler is then paid $\$2^k$ for participating in the game. As the probability of first seeing a head on the kth flip is $1/2^k$, the expected payoff from this game is

$$\begin{aligned} E\left[\text{Payoff}\right] &= \sum_{k \geqslant 1} \frac{1}{2^k} \cdot \$2^k \\ &= \infty. \end{aligned} \tag{C.1}$$

But no gambler would pay an infinite amount to participate in this game. In practice, gamblers seem to be willing to pay no more than about \$20 for the privilege of participating in the St. Petersburg game, a paradox if there ever was one!

Nicolas Bernoulli's cousin Daniel[2] attempted to resolve the paradox by positing that the pleaure (or *utility*) associated with a \$1 increase in wealth would be the reciprocal of the initial level of wealth, and that gamblers could therefore be expected to have logarithmic utility for wealth[3]. The expected utility from participating in the St. Petersburg game is therefore

$$\begin{aligned} E\left[\ln\left(\text{Payoff}\right)\right] &= \sum_{k \geqslant 1} \frac{1}{2^k} \cdot \ln\left(2^k\right) \\ &= \sum_{k \geqslant 1} \frac{k}{2^k} \cdot \ln\left(2\right) \\ &= 2\ln\left(2\right), \end{aligned} \tag{C.2}$$

which is finite.

Moscati (2019) traces the evolution of resolutions to this paradox, and also exposes the vigorous debates between competing schools of thought that finally settled down in the decade following the publication of an axiomatic theory of utility by John von Neumann and Oskar Morgenstern (VNM) in 1947. The VNM axioms are expressed in terms of preferences over random outcomes (or "lotteries"). We write $X \succ Y$ to express a strong preference for lottery X over lottery Y, and $X \sim Y$ to express indifference between the two lotteries. Likewise, we write $X \succeq Y$ to express a weak preference for lottery X over lottery Y. By definition,

[1] Mathematicians and gamblers worked more closely than one might imagine – mathematicians often derived results and sold them to gamblers for immediate use in the field!

[2] The Bernoullis were a remarkable family, as well known for their contributions to mathematics as for their feuds.

[3] As we shall see, these two views are mathematically equivalent. In modern language, Daniel Bernoulli asserted that the marginal utility of wealth was the reciprocal of wealth.

$$X \succeq Y \equiv X \succ Y \text{ or } X \sim Y. \tag{C.3}$$

Axiom 1 (Completeness): All lotteries can be ranked. For any pair of lotteries X and Y, either $X \succ Y$, or $Y \succ X$, or $X \sim Y$.

Axiom 2 (Transitivity): For any three lotteries X, Y and Z, if $X \succeq Y$ and $Y \succeq Z$, then $X \succeq Z$.

Axiom 3 (Substitution): For any three lotteries X, Y and Z, if $X \succeq Y \succeq Z$, then $Y \sim pX + (1-p)Z$ for some $p \in [0,1]$, where $pX + (1-p)Z$ is interpreted as participation in lottery X with probability p and participation in lottery Z with probability $1-p$.

Von Neumann and Morgenstern (1947) proved that if investors' preferences satisfied these three axioms, they would choose among risky alternatives so as to maximize their expected utility, and in spite of many fits, starts, and paradoxes, *Expected Utility Maximization* (EUM) has come to be the dominant paradigm for modeling choice in economics[4].

EUM requires investors to express their preferences via a *cardinal utility function*[5] $U(\mathrm{v})$ that is a dimensionless function of investment end–of–period wealth v, and which is invariant under an affine transformation, $\check{U}(\mathrm{v}) = a + b \cdot U(\mathrm{v}),\ b > 0$, in the sense that $\check{U}$ and $U(\mathrm{v})$ provide the same ranking among investment alternatives.

Suppose an investor's preferences can be captured by such a utility function, whose values are sometimes referred to as *utils*, and she chooses an investment that maximizes expected end–of–period utility $\mathrm{E}[U(\mathrm{V})]$. The form of the utility function U will depend on the assumptions one makes about her preferences, and we next determine the implications of two simple and reasonable assumptions concerning investor preferences for utility functions:

P1: **Non-satiation**: At any level of wealth v_1, the investor prefers a strictly larger wealth $\mathrm{v}_2 > \mathrm{v}_1$.

P2: **Risk Aversion**: The investor maximizes her expected utility, and is *risk averse* in the following sense: she always prefers a fixed end of period wealth, v_0, to a *fair game* in which end of period wealth is the random amount $\mathrm{V} = \mathrm{v}_0 + \mathrm{H}$[6], where H has mean zero and variance $\sigma^2 > 0$.

We note in passing that investors are said to *weakly risk averse* if they are indifferent to participating in such a fair game.[7]

These assumptions are not all–encompassing – the near–universal existence of casinos and lotteries indicates that there is no shortage of individuals who prefer (or sadly, are addicted to) the expected losses and uncertain outcomes that are endemic to gambling. Even more interesting, risk–averse and risk–seeking behavior can co-exist in the same individual: a great many people buy both life insurance and lottery tickets!

An investor with preferences P1 and P2 will use a utility function with the following properties:

U1. $U(\mathrm{v})$ is strictly increasing in v,

[4] Some authors use a more fine-grained set of axioms, see for example the discussions of utility theory in Levy (2016)

[5] Cardinal utility assign a real number to each outcome of interest. Ordinal utility, in contrast, only specifies if one outcome is preferred to another.

[6] This is one of those occasions where it is useful to represent a random variable with a capital letter to distinguish it from an variable that is non–random (such as a particular realization) and so represented by a lower case letter.

[7] Much of the utility theory literature uses the term "strictly risk averse" where we use "risk averse", and uses "risk averse" where we use "weakly risk averse".

U2. $U(\mathrm{v})$ is strictly concave[8].

The utility property U1 follows immediately from P1. The concavity property U2 follows from P2 and Jensen's inequality[9].

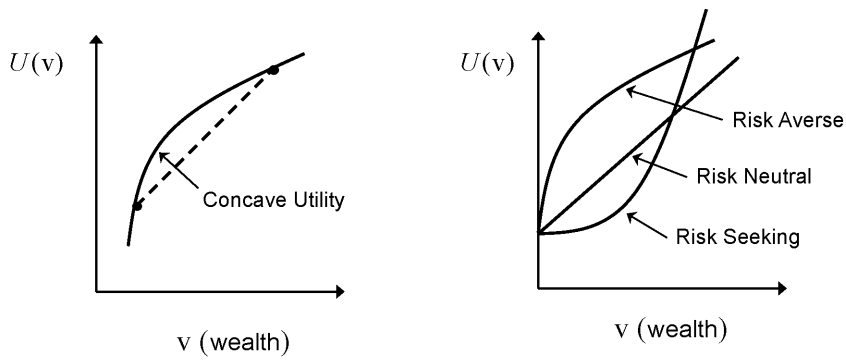

Fig. C.1 Shapes of Utility Functions

The commonly occurring utility functions described in the next subsection have both first and second derivatives, and for such utility functions the properties U1 and U2 are equivalent to the following derivative properties:

U1E: $U'(\mathrm{v}) > 0$

U2E: $U''(\mathrm{v}) < 0$

The first derivative of U is known as its *marginal utility*, and represents the increase in utility obtained from an additional dollar of wealth. While a risk averse EUM investor is one whose differentiable utility function is concave, a *risk seeking* EUM investor is one whose utility function is strictly convex[10]. An investor with a utility function that has a constant first derivative, and hence a zero second derivative, is said to be *risk neutral*.

C.2 Concave Utility Functions and Risk Aversion

With property P2 in mind, consider the fair investment game with end-of-period wealth $\mathrm{V} = \mathrm{v}_0 + \mathrm{H}$, where

$$\mathrm{H} = \begin{cases} \mathrm{h}_1 : \ -\mathrm{v}_0 < \mathrm{h}_1 < 0, & \text{with probability } p \\ \mathrm{h}_2 : \ 0 < \mathrm{h}_2, & \text{with probability } 1-p \end{cases} \tag{C.4}$$

with any fixed h_1 and h_2 such that

[8] A *strictly concave* function is one for which a straight line connecting two distinct points on the $U(\mathrm{v})$ curve lies strictly below the curve, except at the two end-points where the line joins the curve, as in the left–hand plot in Figure C.1. The mathematical definitions of concave and concave functions are given in Appendix A.

[9] See the discussion of convex and concave functions, as well as of Jensen's inequality, in Appendix A

[10] A *strictly convex* utility function is one that satisfies the U2E property with the inequality sign reversed.

$$\mathrm{E}[\mathrm{H}] = p\,\mathrm{h}_1 + (1-p)\,\mathrm{h}_2 = 0, \tag{C.5}$$

so that $\mathrm{E}[\mathrm{V}] = \mathrm{v}_0$.

For this fair game, we have

$$\mathrm{E}[U(\mathrm{V})] = p \cdot U(\mathrm{v}_0 + \mathrm{h}_1) + (1-p) \cdot U(\mathrm{v}_0 + \mathrm{h}_2), \tag{C.6}$$

and since $U(\mathrm{v}_0) = \mathrm{E}[U(\mathrm{v}_0)]$, we also have:

$$\mathrm{E}[U(\mathrm{v}_0)] = U(p \cdot (\mathrm{v}_0 + \mathrm{h}_1) + (1-p) \cdot (\mathrm{v}_0 + \mathrm{h}_2)). \tag{C.7}$$

Since by P2 our EUM investor is risk averse, she prefers the sure outcome v_0 to the risky outcome V, and as she is also an expected utility maximizer, it must be that $\mathrm{E}[U(\mathrm{v}_0)] > \mathrm{E}[U(\mathrm{V})]$, and accordingly we have

$$U(p \cdot (\mathrm{v}_0 + \mathrm{h}_1) + (1-p) \cdot (\mathrm{v}_0 + \mathrm{h}_2)) > p \cdot U(\mathrm{v}_0 + \mathrm{h}_1) + (1-p) \cdot U(\mathrm{v}_0 + \mathrm{h}_2) \tag{C.8}$$

for all combinations of v_0, $-\mathrm{v}_0 < \mathrm{h}_1 < 0$, and $0 < \mathrm{h}_2$ that satisfy (C.5), which is the same as asserting that

$$U(p \cdot \mathrm{v}_1 + (1-p) \cdot \mathrm{v}_2) > p \cdot U(\mathrm{v}_1) + (1-p) \cdot U(\mathrm{v}_2), \ \forall \mathrm{v}_1 < \mathrm{v}_2. \tag{C.9}$$

But the above inequality is equivalent to the statement that U is a strictly concave function.

C.3 Certainty Equivalent Value and the Risk Premium

Sometimes portfolio managers want to compare portfolios that have different expected utilities in terms of monetary units rather the dimensionless units of utility functions. One way in which to do so is to define a dollar amount, v_c, known as the *Certainty Equivalent Value*, that satisfies the equation:

$$\begin{aligned} U(\mathrm{v}_c) &= \mathrm{E}[U(\mathrm{V})] \\ &\implies \mathrm{v}_c = U^{-1}(\mathrm{E}[U(\mathrm{V})]). \end{aligned} \tag{C.10}$$

We interpret v_c as the lower, sure, value that a risk–averse investor would accept in place of the risky payoff V, with higher expected value $\mathrm{E}[\mathrm{V}]$, in order to avoid the uncertainty associated with the random character of V. Figure C.2 shows graphically how such a value is obtained for a risk–averse investor when end–of–period random wealth V has expected utility $\mathrm{E}[U(\mathrm{V})]$. Note that v_c is less than $\mathrm{E}[\mathrm{V}]$, in accordance with Jensen's equality for a (strictly) concave function, which asserts that $\mathrm{E}[U(\mathrm{V})] < U(\mathrm{E}[\mathrm{V}])$ (see Appendix A). Thus

$$U(\mathrm{v}_c) = \mathrm{E}[U(\mathrm{V})] < U(\mathrm{E}[\mathrm{V}])$$

and since U is strictly increasing, it follows that $\mathrm{v}_c < \mathrm{E}[\mathrm{V}]$.

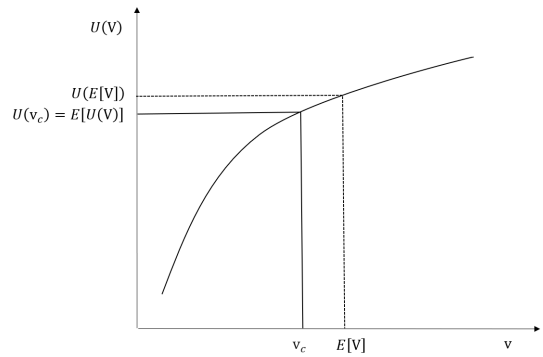

Fig. C.2 Certainty Equivalent Value for Expected Utility

Correspondingly, the *Absolute Risk Premium*, $\pi_A(\mathrm{v})$ is

$$\pi_A(\mathrm{E}[\mathrm{V}]) = \mathrm{E}[\mathrm{V}] - \mathrm{v}_c, \tag{C.11}$$

and this can be interpreted as the amount of wealth the investor is willing to forego (or the insurance premium she is willing to pay) to avoid the risk of a fair investment game. It can also be interpreted as the payment she would require in order to induce her to participate in a fair game.

Closely related to the Absolute Risk Premium is the *Relative Risk Premium*, or the fraction of her wealth that a risk averse investor would pay to avoid the risk of a fair investment game (or would have to be paid in order to induce her to participate in a fair investment game):

$$\begin{aligned}\pi_R(\mathrm{E}[\mathrm{V}]) &= \frac{\pi_A(\mathrm{E}[\mathrm{V}])}{\mathrm{E}[\mathrm{V}]} \\ &= \frac{\mathrm{E}[\mathrm{V}] - \mathrm{v}_c}{\mathrm{E}[\mathrm{V}]}.\end{aligned} \tag{C.12}$$

C.4 Absolute and Relative Risk Aversion

The risk premium is a function both of how uncertain an investment is and the shape of the investor's utility curve, and we so separate the contributions of these two influences. Once again, let v_0 be an initial level of wealth, let $\mathrm{V} = \mathrm{v}_0 + \mathrm{H}$ with random payoff H be the end-of-period random wealth, and let $U(\mathrm{V})$ be the investor's utility function. Suppose further that the investment is a fair game with $\mathrm{E}[\mathrm{H}] = 0$, and let σ_{V}^2 be the variance of H. Let v_c be the certainty equivalent for $\mathrm{E}[\mathrm{V}]$ and $\pi_A(\mathrm{V})$ be the associated absolute risk premium.

If we expand $U(\mathrm{V})$ in a two–term Taylor series around its mean, we get

$$\begin{aligned}U(\mathrm{V}) &\approx U(\mathrm{E}[\mathrm{V}]) + U'(\mathrm{E}[\mathrm{V}])(\mathrm{V} - \mathrm{E}[\mathrm{V}]) + \frac{1}{2} \cdot U''(\mathrm{V})(\mathrm{V} - \mathrm{E}[\mathrm{V}])^2 \\ &= U(\mathrm{v}_0) + U'(\mathrm{v}_0)(\mathrm{V} - \mathrm{v}_0) + \frac{1}{2} \cdot U''(\mathrm{v}_0)(\mathrm{V} - \mathrm{v}_0)^2\end{aligned} \tag{C.13}$$

where the second line follows from the observation that $\mathrm{E}[\mathrm{V}] = \mathrm{v}_0 + \mathrm{E}[\mathrm{H}] = \mathrm{v}_0$.

Likewise, we can perform a one-term Taylor series expansion of $U(\mathrm{V})$ around its mean and write

$$\begin{aligned} U(\mathrm{v_c}) &\approx U(\mathrm{E}[\mathrm{V}]) + U'(\mathrm{E}[\mathrm{V}])(\mathrm{v_c} - \mathrm{E}[\mathrm{V}]) \\ &= U(\mathrm{v}_0) + U'(\mathrm{v}_0)(\mathrm{v_c} - \mathrm{v}_0). \end{aligned} \tag{C.14}$$

On taking the expectation of both sides of (C.13), we see that

$$\begin{aligned} \mathrm{E}[U(\mathrm{V})] &\triangleq U(\mathrm{v}_c) \\ &\approx U(\mathrm{v}_0) + \frac{1}{2} \cdot U''(\mathrm{v}_0) \cdot \sigma_H^2 \end{aligned} \tag{C.15}$$

Equating (C.14) and (C.15) and simplifying, we get

$$\begin{aligned} \mathrm{v}_0 - \mathrm{v_c} &\triangleq \pi_A(\mathrm{v}_0) \\ &= -\frac{1}{2} \cdot \frac{U''(\mathrm{v}_0)}{U'(\mathrm{v}_0)} \cdot \sigma_H^2 \\ &= \frac{1}{2} \cdot ARA(\mathrm{v}_0) \cdot \sigma_H^2, \end{aligned} \tag{C.16}$$

where Absolute Risk Aversion is defined to be

$$ARA(\mathrm{v}) = -\frac{U''(\mathrm{v})}{U'(\mathrm{v})}. \tag{C.17}$$

We can derive a similar expression for Relative Risk Aversion starting with (C.12):

$$\begin{aligned} \pi_R(\mathrm{v}_0) &= \frac{\pi_A(\mathrm{v}_0)}{\mathrm{v}_0} \\ &= -\frac{1}{2} \cdot \frac{U''(\mathrm{v}_0)}{U'(\mathrm{v}_0)} \cdot \frac{\mathrm{v}_0}{\mathrm{v}_0^2} \cdot \sigma_H^2 \\ &= -\frac{1}{2} \cdot \frac{U''(\mathrm{v}_0) \cdot \mathrm{v}_0}{U'(\mathrm{v}_0)} \cdot \sigma_{H/\mathrm{v}_0}^2 \\ &= \frac{1}{2} \cdot RRA(\mathrm{v}_0) \cdot \sigma_{H/\mathrm{v}_0}^2 \end{aligned} \tag{C.18}$$

where Relative Risk Aversion (RRA) is defined as:

$$\begin{aligned} RRA(\mathrm{v}) &= \mathrm{v} \cdot ARA(\mathrm{v}) \\ &= -\frac{\mathrm{v} \cdot U''(\mathrm{v})}{U'(\mathrm{v})}. \end{aligned} \tag{C.19}$$

Given the relationship between ARA and RRA functions in (C.19), it is clear that a constant relative risk aversion (CRRA) investor must exhibit decreasing ARA.

Both ARA and RRA characterize utility functions locally, but it is their global behavior for all possible values of v that makes them useful in practice. Empirical evidence suggests that most investors exhibit decreasing absolute risk aversion (i.e. they are more willing to risk a fixed amount of money as their wealth increases), but Relative Risk Aversion is known to depend in a complex way on age, sex, and wealth, among other characteristics (see Bellante and Greene (2004) and Halek and Eisenhauer (2001) for good discussions of this important point). It is therefore common practice to simply assume that investors exhibit Constant Relative Risk Aversion (i.e. to assume that the fraction of their wealth that they are willing to risk is independent of its level).

Marginal Utility and Risk Aversion of Common Utility Functions

We next explore the marginal utility, as well as the absolute and relative risk aversions, of some common utility functions. As risk aversion is rooted both in human psychology and in our environment[11], it is not easy to divine the true shape of an investor's utility function or degree of risk aversion. It has therefore become common practice to focus attention on utility functions that are analytically tractable, or which correspond to easily visualized economic goals.

Logarithmic (or Log) Utility

For log utility

$$U(\mathrm{v}) = \ln(\mathrm{v}),\ \mathrm{v} > 0, \tag{C.20}$$

so that $U'(\mathrm{v}) = 1/\mathrm{v}$ and $U''(\mathrm{v}) = -1/\mathrm{v}^2$ for positive v, and from (C.17) and (C.19) we get:

$$ARA(\mathrm{v}) = \frac{1}{\mathrm{v}}, \tag{C.21a}$$

$$RRA(\mathrm{v}) = 1. \tag{C.21b}$$

The marginal utility of wealth, $U'(\mathrm{v})$, is $1/\mathrm{v}$, just as Daniel Bernoulli posited. Logarithmic utility plays an important role in finance, as it is closely related to the geometric mean, and we explore the link between the two in Section C.8.

Generalized Logarithmic (or GLUM) Utility

Generalized logarithmic utility is an exceptionally powerful and underappreciated variant of log utility. It was introduced in Rubinstein (1976) and Rubinstein (1977) as a flexible alternative to the CAPM, but did not achieve widespread popularity. It allows for a wide range of investor preferences, and lends itself naturally to the management of assets and liabilities, as both are easily interpreted as terms in the utility function. Formally

[11] Investors often become more risk seeking in good times (such as the run up to the Internet bubble in the late 1990s) and more risk averse in bad times (such as the period immediately following the start of the Great Financial Crisis in 2008).

$$U(\mathrm{v}) = \ln(\mathrm{v} + A),\ \mathrm{v} > -A, \tag{C.22}$$

so that $U'(\mathrm{v}) = 1/(\mathrm{v}+A)$ and $U''(\mathrm{v}) = -1/(\mathrm{v}+A)^2$ for $\mathrm{v} > -A$, and from (C.17) and (C.19) we get:

$$ARA(\mathrm{v}) = \frac{1}{\mathrm{v}+A}, \tag{C.23a}$$

$$RRA(\mathrm{v}) = \frac{\mathrm{v}}{\mathrm{v}+A}. \tag{C.23b}$$

The marginal utility of wealth, $U'(\mathrm{v})$, is clearly decreasing in wealth, as is Absolute Risk Aversion. More importantly, Relative Risk Aversion is increasing, constant, or decreasing in wealth according to whether A is positive, zero, or negative, allowing a wide range of investor preferences to be parsimoniously modeled. If we interpret $-A$ as the present value of the investor's liabilities, $\mathrm{v}+A$ is her *actuarial surplus*, or *discretionary wealth*: she could, in principle at least, lose all of it, and still be able to meet her spending commitments. This is equivalent to the requirement that $\mathrm{v}+A > 0$.The Generalized Logarithmic Utility function is therefore of great importance when modeling the wealth of households, and of insurers, who have large portfolios of assets that are offset by similar large portfolios of liabilities.

Generalized logarithmic utility is sometimes expressed in terms of returns as

$$U(r) = \frac{\ln(1+\lambda r)}{\lambda},\ r > -\frac{1}{\lambda},\ \lambda > 0, \tag{C.24}$$

and expanding (C.24) in a two-term Taylor series around $r = 0$ gives

$$U(r) \approx r - \frac{\lambda}{2} r^2. \tag{C.25}$$

Taking expecations on both sides of (C.24) gives

$$\begin{aligned} E[U(r)] &\approx E[r] - \frac{\lambda}{2}\left(\sigma^2 + E[r]^2\right) \\ &\approx E[r] - \frac{\lambda}{2}\sigma^2 \end{aligned} \tag{C.26}$$

which closely mimics the form of the Quadratic Utility function described in Section 2.5, allowing us to identify λ as a measure of risk aversion.

Wilcox, Zieff, et al. (2025) use (C.24) and (C.26) to allow the identification of λ by asking an investor a single question: "What is f_{max}, the largest fraction of your *financial* wealth you can lose without it ruining your life?" Viewed through the lens of (C.24), this is equivalent to identifying the most negative return $r_{max} = -f_{max}$ that the investor can tolerate; at this return, her financial wealth is reduced to 0 and her utility becomes infinitely negative.

Wilcox, Zieff, et al. (2025) argue that it proves effective in practice to set $\lambda = 1/f_{max}$. If the investor can lose all her financial wealth, $\lambda = 1$, and she will invest log-optimally. Many investors say that they can lose between one-quarter and one-half of their financial wealth, corresponding to values of λ between 2 and 4, which corresponds remarkably well to the range of λ obtained through the use of more complex and comprehensive questionnaires.

Power Utility

A power utility function has the form

$$U(\mathrm{v};\gamma) = \frac{\mathrm{v}^{1-\gamma}}{1-\gamma}, \quad \mathrm{v} > 0, \quad \gamma > 0, \quad \gamma \neq 1 \tag{C.27}$$

It is an easy exercise to show that power utility has the risk aversion functions:

$$ARA(\mathrm{v}) = \frac{\gamma}{\mathrm{v}}, \tag{C.28a}$$

$$RRA(\mathrm{v}) = \gamma. \tag{C.28b}$$

and marginal utility

$$U'(\mathrm{v}) = \mathrm{v}^{-\gamma}. \tag{C.29}$$

As $\gamma \longrightarrow 1$, power utility converges to logarithmic utility, and this is easily seen by applying L'Hospital's rule to (C.27). Investors who are attracted to the CRRA and long run growth properties of logarithmic utility, but who find it too risky for their taste, will find that power utility offers much of what logarithmic utility does, but with the ability to control risk by making an appropriate choice for γ.

Exponential Utility

The exponential utility function has the form

$$U(\mathrm{v}) = \frac{1-\exp(-c\cdot\mathrm{v})}{c}, c > 0, \mathrm{v} \geqslant 0, \tag{C.30}$$

with c being a constant, so that its marginal utility is $\exp(-c\cdot\mathrm{v})$, and its risk aversion functions are

$$ARA(\mathrm{v}) = c \tag{C.31a}$$

$$RRA(\mathrm{v}) = c\mathrm{v}. \tag{C.31b}$$

Even though these do not correspond to commonly accepted views on investor risk aversions, exponential utility is sometimes used in economics on account of its analytic tractability.

Quadratic Utility Functions

A quadratic utility function has the form

$$U(\mathrm{v}) = \mathrm{v} - c\mathrm{v}^2, c > 0, \mathrm{v} > 0, \tag{C.32}$$

with c being a constant, so that its marginal utility is $1-2c\mathrm{v}$, and its risk aversion functions are

$$ARA(\mathrm{v}) = \frac{2c}{1-2c\mathrm{v}} \tag{C.33a}$$

$$RRA(\mathrm{v}) = \frac{2c\mathrm{v}}{1-2c\mathrm{v}}. \tag{C.33b}$$

But these risk aversion functions are infinite for $\mathrm{v} = 1/2b$ and negative for $\mathrm{v} > 1/2b$, and the quadratic utility function is therefore useless for such values of v. In spite of these severe limitations, quadratic utility plays an important role in portfolio management because most any utility function can be approximated reasonably well by a two-term Taylor series expansion around the current level of wealth.

All these five utility functions belong to the class of Hyperbolic Absolute Risk Aversion (HARA) utility functions, which are characterized by the fact that their reciprocals are linear functions of wealth, i.e.

$$ARA(\mathrm{v}) = \frac{1}{\gamma\mathrm{v}+\delta}. \tag{C.34}$$

HARA utility functions are often used on account of their analytic tractability, even when a careful analysis of their properties suggests caution with their use. Table C.1 displays the risk aversion properties of these five utility functions, where these properties for quadratic utility only hold for $\mathrm{v} < 1/2c$.

In our view, the first three utility functions best model rational human economic behavior, while quadratic utility is a local approximation that works surprisingly well in practice for a variety of problems in portfolio management, even though it was characterized by Arrow (1971) as "absurd". As for exponential utility, it is egregious in failing to reflect typical investor behavior with respect to both ARA and RRA, but it sometimes used to justify mean–variance analysis for normally distributed security returns.

Table C.1 Absolute and Relative Risk Aversion for Various Utility Functions

Utility Function	Absolute and Relative Risk Aversion
Logarithmic	Decreasing ARA and constant RRA
Generalized Logarithmic	Decreasing ARA and increasing, constant, or decreasing RRA
Power	Decreasing ARA and constant RRA
Exponential	Constant ARA and linearly increasing RRA
Quadratic	Increasing ARA and increasing RRA

The results in Table C.1 show that a CRRA investor would use a log or power utility function, and would not use a quadratic or exponential utility function.

C.5 Expected Utility Maximization and Mean Variance Optimization

EUM with Certain Families of Returns Distributions

If asset returns are multivariate normal, then portfolio return $r_P = \mathbf{w}'\mathbf{r}$ has a normal distribution with a mean $\mu_P = \mathbf{w}'\boldsymbol{\mu}$ and variance $\sigma_P^2 = \mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}$. In this case the expected utility $\mathrm{E}[U(\mathrm{v}_0 \cdot (1+r_P))]$ will be

a deterministic function $g(\mu_P, \sigma_P)$ of the portfolio's mean return and standard deviation. When U is strictly increasing, it can be shown that $g(\mu_P, \sigma_P)$ is increasing in μ_P and decreasing in σ_P, and the EUM investor holds MinVar portfolios[12].

For the sake of having a mathematical example where EUM does lead directly to mean-variance optimization, some quantitative finance papers and books discuss the fact that if an exponential utility function is used, and portfolio returns are normally distributed, then EUM leads to the quadratic utility form of MVO. But in this case, we have not only the weakness of the normal portfolio returns assumption, but also the inappropriateness of the exponential utility function due to its constant ARA and increasing RRA.

Beneveniste et al. (2024) provide a more robust defence of mean-variance optimization for a class of *standard utility functions* that are increasing, strictly concave and continuously differentiable. They prove that utility maximization for any standard utility function is equivalent to mean-variance optimization for some level of risk aversion if and only if the security returns vector $\mathbf{r}$ admits the representation

$$\mathbf{r} = \boldsymbol{\mu} Z + \epsilon, \tag{C.35}$$

where $\boldsymbol{\mu}$ is a vector, Z is a scalar random variable with $E[Z] = 1$, and ϵ is a random vector with

$$E\left[\epsilon' \boldsymbol{\sigma}^{-1} \boldsymbol{\mu}\right] = 0 \tag{C.36a}$$

$$E[\epsilon | Z] = \mathbf{0} \tag{C.36b}$$

$$\mathbf{C}_\epsilon = \boldsymbol{\sigma} - var(Z)\, \boldsymbol{\mu}\boldsymbol{\mu}' \tag{C.36c}$$

for some Positive Semi-Definite covariance matrix $\boldsymbol{\sigma} = \mathbf{C}_\mathbf{r}$.

An empirical justification for quadratic utility is provided by the frequently referenced study by Levy and Markowitz (1979a), who compared the performance of log utility with that of an approximation based on quadratic utility, using actual returns for 149 investment companies from 1958 to 1967. While their conclusions indicate that quadratic utility provides a reasonable approximation to log utility, the results are based on a limited data set and must be applied with care to modern markets in which a wide range of derivatives are traded. Consequently, their results, while useful, do not provide convincing evidence for the adequacy of the quadratic utility approximation to expected log utility in the context of modern financial markets.[13]

Additionally, security returns do not follow a multivariate normal distribution: there is considerable empirical evidence that such an assumption is not justified for the majority of assets. For example, the returns of small-cap stocks and hedge fund returns are often highly non-normal, exhibiting both skewness and kurtosis. Elliptical distributions are somewhat more promising, but their elliptical contours of constant probability also often fail to adequately capture the reality of multivariate returns distributions that occur in practice.

Last, but not least, one should keep in mind that since arithmetic returns are bounded below by −100%, no standard distribution (e.g. normal distribution, t– or skew–t distribution) that assigns positive probability to all intervals will be entirely adequate. One way in which to address this problem is to use log–normal returns, which are bounded below by −100% and which compound correctly over time,

[12] The necessary details are provided in Chapter 4 of Ingersoll (1987).

[13] Kroll et al. (1984) expands on Levy and Markowitz (1979b) – it considers a wide range of utility functions, and also explores the use of individual securities in place of funds, but, like its predecessor, does not explore the use of options or other derivatives.

but which do not aggregate to a log–normal distribution when weighted and aggregated across portfolio assets.

Expected Utility Taylor Series Approximation and Quadratic Utility

A common alternative approach to assuming portfolio normal returns is to assume that the investor uses a constant relative risk aversion (CRRA) utility function, and use a two-term plus remainder Taylor series expansion of the utility function to show that EUM leads to MVO portfolios. In doing so, one needs to decide what value to do the Taylor series expansion about, and the two natural choices are either to expand expected utility about portfolio mean return μ_P or expand it about $r_P = 0$. Using the latter choice gives

$$U(\mathrm{v}_0 \cdot (1 + r_P)) = U(\mathrm{v}_0) + U'(\mathrm{v}_0) \cdot r_P + \frac{1}{2} U''(\mathrm{v}_0 \cdot (1 + \mu_P)) \cdot r_P^2 + HOT \tag{C.37}$$

where the higher-order terms HOT depend on the higher order moments of r_P. Taking expectations gives

$$\begin{aligned} \mathrm{E}\left[U(\mathrm{v}_0 \cdot (1 + r_P))\right] &= U(\mathrm{v}_0) + U'(\mathrm{v}_0) \cdot \mathrm{E}\left[r_P\right] + \frac{1}{2} U''(\mathrm{v}_0) \cdot \mathrm{E}\left[r_P^2\right] + \mathrm{E}\left(HOT\right) \\ &= U(\mathrm{v}_0) + U'(\mathrm{v}_0) \cdot \mu_P + \frac{1}{2} U''(\mathrm{v}_0) \cdot \left(\sigma_P^2 + \mu_P^2\right) + \mathrm{E}\left(HOT\right) \end{aligned} \tag{C.38}$$

Ignoring the expected value of the higher order terms, and recalling that the argument of the utility function on the left hand side is the end of period portfolio value V, we have the approximation

$$\begin{aligned} \mathrm{E}\left[U(\mathrm{V})\right] &\approx U(\mathrm{v}_0) + U'(\mathrm{v}_0) \left(\mu_P + \frac{1}{2} \cdot \frac{U''(\mathrm{v}_0)}{U'(\mathrm{v}_0)} \cdot \left(\sigma_P^2 + \mu_P^2\right)\right) \\ &= U(\mathrm{v}_0) + U'(\mathrm{v}_0) \left(\mu_P - \frac{1}{2} ARA_U(\mathrm{v}_0) \cdot \left(\sigma_P^2 + \mu_P^2\right)\right) \end{aligned} \tag{C.39}$$

where $ARA_U(\mathrm{v}_0)$ is the absolute risk aversion function for the utility function U.

So long as $U'(\mathrm{v}_0)$ is positive, which will be the case for any strictly increasing utility function, maximizing $\mathrm{E}\left[U(\mathrm{V})\right]$ is equivalent to maximizing the mean-variance objective function with a special value of the risk aversion parameter that is determined by the utility function

$$Q_U(\mu_P, \sigma_P) = \mu_P - \frac{1}{2} ARA_U(\mathrm{v}_0) \cdot \left(\sigma_P^2 + \mu_P^2\right). \tag{C.40}$$

Note that the above mean-variance objective function coincides with the quadratic utility functions of Section 2.3 only if, in addition to replacing $ARA(\mathrm{v}_0)$ above by the general risk aversion parameter λ, one assumes (as is often the case) that the portfolio mean return is so small relative to the portfolio standard deviation that the μ_P^2 can be dropped in the expression for $Q_U(\mu_P, \sigma_P)$. In this case we are back to the quadratic utility version of MVO discussed extensively in that section.

However, the dropping of the expected values of the higher order terms in (C.37) is in general questionable in view of the empirical evidence that asset returns are non–normally distributed, and often

exhibit skewness and fat tails to a greater or lesser degree, particularly in the case of small-cap and micro-cap stocks, and in hedge fund returns.

Finally, we mention that the Taylor series expansion of a utility function about a portfolio mean return μ_P results in an expression that depends on μ_P and σ_P differently than (C.39), but which in general depends on μ_P in a utility function dependent manner.

C.6 Expected Quadratic Utility and QP Optimization

We note that direct use of a quadratic utility function of wealth, expressed via portfolio return gives:

$$\begin{aligned}
\mathrm{E}\left[U_{quadratic}(\mathrm{V})\right] =& \mathrm{E}\left[\mathrm{v}_0\,(1+r_P) - b\,\mathrm{v}_0^2\,(1+r_P)^2\right] \\
=& \mathrm{v}_0\,\mathrm{E}\left[1 + r_P - b\cdot\mathrm{v}_0\,(1+2r_P+r_P^2)\right] \\
=& \mathrm{v}_0\,(1-\mathrm{v}_0\,b) + (1-2\,b\,\mathrm{v}_0)\,\mathrm{E}\left[r_P\right] - b\,\mathrm{v}_0\,\mathrm{E}\left[r_P^2\right] \\
=& \mathrm{v}_0\,(1-\mathrm{v}_0\,b) + (1-2\,b\,\mathrm{v}_0)\left[\mu_P - \frac{b\,\mathrm{v}_0}{1-2\,b\,\mathrm{v}_0}\,(\sigma_P^2+\mu_P^2)\right].
\end{aligned} \tag{C.41}$$

For $\mathrm{v}_0 < 2b$, maximization of $\mathrm{E}\left[U(\mathrm{V})\right]$ is equivalent to maximization of the quantity in the square bracket above, which is not surprisingly a special case of (C.39) with U being the quadratic utility function.

Expected Quadratic Utility QP Problem

Re-writing (C.40) with the ARA factor replaced by a general risk aversion parameter λ gives

$$\mathrm{Q}_U(\mu_P,\sigma_P) = \mu_P - \frac{1}{2}\lambda\left(\sigma_P^2+\mu_P^2\right) \tag{C.42}$$

which in terms of $\mu_P = \mathbf{w}'\boldsymbol{\mu}$ and $\sigma_P^2 = \mathbf{w}'\mathbf{C}\mathbf{w}$, with $\mathbf{C}$ being the returns covariance matrix, is:

$$\begin{aligned}
\mathrm{Q}_U(\mu_P,\sigma_P) &= \mathbf{w}'\boldsymbol{\mu} - \lambda\cdot(\mathbf{w}'\mathbf{C}\mathbf{w} + \mathbf{w}'\boldsymbol{\mu}\boldsymbol{\mu}'\mathbf{w}) \\
&= \mathbf{w}'\boldsymbol{\mu} - \lambda\cdot w'\check{\mathbf{C}}\mathbf{w}
\end{aligned}$$

where $\check{\mathbf{C}} = \mathbf{C} + \boldsymbol{\mu}\boldsymbol{\mu}' = E\left[\mathbf{r}\mathbf{r}'\right]$.

In other words, the use of a quadratic utility function in the EUM framework leads to an objective function that differs from the standard QU objective function only by virtue of replacing the covariance matrix $\mathbf{C}$ in the latter with the second moment matrix $\check{\mathbf{C}} = \mathrm{E}\left[\mathbf{r}\mathbf{r}'\right]$. Keeping in mind the fact that many practitioners estimate the returns covariance by assuming zero mean for the returns, this is perhaps not a significant difference between maximizing the expected value of quadratic utility and use of the QU optimization form of MVO.

C.7 Objections to Utility Theory

In spite of its impeccable pedigree, EUM has long been plagued by paradoxes and inconsistent choices made by participants in controlled laboratory experiments. These experiments are most often expressed in the form of choices between lotteries that look superficially different, but which, upon reflection, prove to be similar. Alchian (1953) offers the following example, which he attributes to Markowitz.

A friend offers you the choice of participation in one of three lotteries:

- Lottery A consists of 2,000 tickets, two of which are marked \$1,000 and the rest are marked \$0.
- Lottery B consists of 2,000 tickets, twenty of which are marked \$100 and the rest are marked \$0.
- Lottery C consists of 2,000 tickets, one of which is marked \$1,000, ten of which are marked \$100, and the rest are marked \$0.

How would you rank these lotteries, and why?

A simple calculation shows that lottery C can be synthesized by flipping a fair coin that forces participation in lottery A or lottery B, each with a probability of 50%. Under the VNM axioms, it cannot therefore be rationally preferred to both lotteries A and B. In practice, however, a surprising number of people who are presented with this choice rank lottery C either first or third among these three lotteries! It is an easy matter to determine that the expected payoff from each of these lotteries is \$1, so that the error is not driven by their expected payoffs.

Chapter 15 of Levy (2016) documents a variety of similar paradoxes due to Allais (1953), Ellsberg (1961), and others, in which volunteers systematically make choices that violate the EUM paradigm. Many, but not all, of these paradoxes are driven by the presence of rare outcomes which prove disproportionately attractive or repellent to participants, or by a presentation that disguises the common aspects of seemingly different choices; but some, such as Ellsberg's paradox, are also driven by a human aversion to ambiguity.

Ellsberg's paradox can be illustrated by a simple coin tossing game in which the coin that is tossed can be chosen by the participant. The first coin that is offered to the participant is known to be fair, i.e

$$p_{Head,1} = p_{Tail,1} = \frac{1}{2}. \tag{C.43}$$

The second coin is known to be fair in expectation – the probability that it comes up heads not known to the participant, who knows only that it is uniformly distributed on $[0,1]$. It is easily seen that

$$\mathrm{E}\left[p_{Head,2}\right] = \mathrm{E}\left[p_{Tail,2}\right] = \frac{1}{2}. \tag{C.44}$$

Regardless of which coin she chooses, she wins \$1 if the coin comes up heads, and loses \$1 if it comes up tails. The expected value of both games is 0, and their expected utilities are identical for any level of initial wealth, and for any well–defined utility function, but individuals who are offered this choice overwhelmingly choose the first coin, and exhibit a deep aversion to the uncertainty associated with the second coin.

No perfect resolution to these paradoxes exist, as they appear to be deeply rooted in the way the human mind works[14]. In early work, Markowitz (1952) suggests that utility would be determined by a change in,

[14] Kahneman and Tversky (2011) offers a comprehensive and readable account of many of these paradoxes and their relationship to the way in which our minds work (or are currently thought to work).

rather than the absolute level of, an investor's wealth, and it is worth re–examining the St. Petersburg Paradox in this light.

Consider an investor with initial wealth v_0, and logarithmic utility function $U(\mathrm{v}) = \ln(\mathrm{v})$. The change in her expected utility from playing the St. Petersburg game without paying any entrance fee is:

$$\begin{aligned}\Delta U_{Free}(\mathrm{v}_0) &= \mathrm{E}[U(\mathrm{V})] - U(\mathrm{v}_0) \\ &= \sum_{k \geqslant 1} \frac{1}{2^k} \cdot \ln\left(\mathrm{v}_0 + 2^k\right) - U(\mathrm{v}_0).\end{aligned} \tag{C.45}$$

If, on the other hand, she must pay a fixed amount, say $\$P(\mathrm{v}_0)$, for the privilege of playing the game, the change in her expected utility is

$$\begin{aligned}\Delta U_{Paid}(\mathrm{v}_0) &= \mathrm{E}[U(\mathrm{V})] - U(\mathrm{v}_0) \\ &= \sum_{k \geqslant 1} \frac{1}{2^k} \cdot \ln\left(\mathrm{v}_0 + 2^k - P(\mathrm{v}_0)\right) - U(\mathrm{v}_0),\end{aligned} \tag{C.46}$$

and we can use search techniques to determine the value of $P(\mathrm{v}_0)$ that forces $\Delta U_{Paid}(\mathrm{v}_0)$ to equal 0 (i.e. for no change in utility) for any value of v_0. Table C.2 displays v_0, $\Delta U_{Free}(\mathrm{v}_0)$, $P(\mathrm{v}_0)$ and $P(\mathrm{v}_0)/\mathrm{v}_0$ for values of v_0 ranging from \$1 to \$8,192.

Table C.2 The St. Petersburg Game With Varying Initial Levels of Wealth

v_0	$\Delta U_{Free}(\mathrm{v}_0)$	$P(\mathrm{v}_0)$	$\frac{P(\mathrm{v}_0)}{\mathrm{v}_0}$
\$1	1.6646	2.8160	2.8160
\$2	1.1789	3.3476	1.6738
\$4	0.7922	4.0000	1.0000
\$8	0.5077	4.7245	0.5906
\$16	0.3127	5.5052	0.3441
\$32	0.1867	6.3360	0.1980
\$64	0.1087	7.2100	0.1127
\$128	0.0621	8.1196	0.0634
\$256	0.0349	9.0569	0.0354
\$512	0.0194	10.0148	0.0196
\$1,024	0.0107	10.9873	0.0107
\$2,048	0.0058	11.9698	0.0058
\$4,096	0.0032	12.9589	0.0032
\$8,192	0.0017	13.9522	0.0017

As wealth increases, the incremental utility gained by participation in the game becomes vanishingly small, and the maximum price that an investor would pay for the privilege of participation grows slowly (approximately logarithmically) with wealth. Strikingly, a participant with a wealth of \$4 would be willing to risk *all* her wealth on the St. Petersburg game, while one with a wealth of \$1 would actually borrow \$1.816 and pay \$2.816 for the privilege of participating! At the other extreme, an investor with

$8,192 would pay only $13.95 to participate in this game. The results in Table C.2 accord reasonably well with the results of experiments conducted with actual subjects with varying degrees of wealth. Also of interest is the fact that the additional utility gained by participating in the St. Petersburg game converges to the fraction of her wealth that she would pay for the privilege of participating.

Markowitz's insight that utility should be driven by changes in wealth, and not by the level of wealth, is the starting point for a variety of proposed resolutions to these paradoxes. *Prospect Theory*, introduced in Kahneman and Tversky (1979), proposes that at any level of wealth, investors will evaluate lotteries through the lens of a sharply kinked and S–shaped *Value function* that is gently increasing (and concave) in gains and sharply decreasing (and convex) in losses, both measured from the current level of wealth. Almost contemporaneously, Loomes and Sudgen (1982) proposed *Regret Theory*, which constructs a modified utility function that captures the regret or rejoicing that an investor might feel on account of both the choices they make and the results they achieve. Bleichrodt and Wakker (2015) provide an excellent retrospective on Regret Theory and compare the insights in Loomes and Sudgen (1982) to those in Kahneman and Tversky (1979).

Levy and Kroll (1976) and Levy (2016) propose stochastic dominance as the appropriate lens through which to compare investments, and Tversky and Kahneman (1992) present an updated version of their theory, now called *Cumulative Prospect Theory*, to better accommodates multiple outcomes and also to correct some of its failures that relate to stochastic dominance.

While these approaches have all attracted significant attention, and almost universally argue that the VNM axioms are an overly restrictive definition of rational behavior, they have not had a significant practical impact on portfolio construction for a variety of reasons, including:

1. Most investors cannot specify their personal utility function with any degree of precision,
2. These theories tend to focus on lotteries with a few discrete outcomes, while security returns are a continuous random variable,
3. The joint distribution of security returns is not stable – it exhibits significant time variation, and is not easily modeled, complicating the computation of expected utility, and
4. The distribution of a portfolio's returns must be obtained from that of its constituents by evaluating a high–order convolution integral over the joint distribution of individual security returns, again complicating the computation of expected utility.

Thus, in spite of its enormous clarifying impact on the the underpinnings of portfolio theory, utility theory has not transformed the practice of portfolio management, and, in the next section, we discuss how some of its insights can be used in practice to good effect.

C.8 Making Utility Theory Practical

The biggest single stumbling block to the use of utility theory in portfolio construction is the almost universal inability to specify one's utility function. We therefore suggest focusing on a few utility functions that translate naturally to reasonable economic objectives.

We recommend thinking about utility functions in terms of their Relative Risk Aversion, and not their Absolute Risk Aversion, because most investors think naturally in terms of return and risk, and not in terms of lotteries. Given that the expected returns of most asset classes are not large, an investor's wealth will not vary significantly from one period to the next, and it seems to us that it most natural to focus on utility functions with Constant Relative Risk Aversion, particularly logarithmic and power

utility. When one wishes to model assets as well as liabilities, Generalized Logarithmic utility is almost always the best choice, as the constant A can be thought of as the negative of the present value of one's liabilities.

There is a rich literature (and a correspondingly angry debate) on log utility and log–optimal investment, as maximizing log utility is equivalent to maximizing geometric return, and thus to maximizing terminal wealth. Maclean et al. (2010) provide a concise introduction to the theory and practice of log–optimal investment, as well as useful guide to the controversies that surround it. To expose the central idea, we let $r_1, r_2, \cdots, r_T$ be the return of a portfolio with an initial value of v_0. The portfolio's terminal value, V_T is given by

$$\begin{aligned}\mathrm{V}_T &= \mathrm{v}_0 \cdot \prod_{i=1}^{T} (1 + r_i) \\ &= \mathrm{v}_0 \cdot (1 + g)^T \end{aligned} \tag{C.47}$$

where g, the geometric return, is given by

$$g = \left(\prod_{i=1}^{T} (1 + r_i) \right)^{\frac{1}{T}} - 1. \tag{C.48}$$

We can also write (C.47) as

$$\begin{aligned}\mathrm{V}_T &= \mathrm{v}_0 \cdot \exp\left(\sum_{i=1}^{T} \ln(1 + r_i) \right) \\ &= \mathrm{v}_0 \cdot \exp\left(\sum_{i=1}^{T} \widetilde{r}_i \right) \\ &= \mathrm{v}_0 \cdot \exp\left(T \cdot \frac{1}{T} \sum_{i=1}^{T} \widetilde{r}_i \right) \\ &\longrightarrow \mathrm{v}_0 \cdot \exp\left(T \cdot \mathrm{E}\left[\widetilde{r}\right] \right), \end{aligned} \tag{C.49}$$

where the last step rests on the assumption that the distribution of returns is stable over time. As e^x is monotonically increasing in x, it is clear that maximizing terminal wealth is equivalent to maximizing expected logarithmic return, or equivalently, to maximizing expected log utility for the portfolio.

In Chapter 3, it is shown that

$$g \approx \mu_r - \frac{1}{2} \sigma_r^2, \tag{C.50}$$

so that maximizing g is approximately equivalent to a mean–variance optimization with $\lambda = 1$.

If this level of relative risk aversion seems too low, we can explore power utility functions with higher levels of risk aversion, and experiments have shown that RRA lies between 2 and 4 for a large fraction of the population. Having chosen a utility function, we proceed in one of two ways:

1. Set the parameters of the chosen utility function to match the investor's RRA (this is often determined via a questionnaire or by analyzing her choices in one or more lotteries), and construct mean–variance optimal portfolios with the same level of risk aversion. In effect, we construct a local approximation to our preferred utility function, allowing the entire machinery of mean–variance analysis to be brought to bear on the problem.
2. Insert the chosen utility function into a simulation and historical back–testing engine, and then identify the utility maximizing optimal portfolio using sampling or some form or gradient search. Adler and Kritzman (2007) and Sharpe (2007) describe simple approaches to utility maximization in an asset–only setting, and Wilcox (2019) and Wilcox and Satchell (2021) show how to optimize the growth of actual surplus using Generalized Logarithmic Utility.

Each method has its advantages, though satisfactory results can be obtained from either one when simple securities with relatively little skew or kurtosis are the only financial instruments used. When securities with skewed and fat–tailed distributions, such as small–cap stocks, high–yield bonds, options and other non–linear derivatives, are used, the second method dominates.

Regardless of the method used, it is vital to start with good, forward–looking, estimates of expected returns of the various markets in which one can invest and to ensure that one's estimates of individual security returns and volatilities are consistent with these estimates[15], as estimation errors can cause extreme instability in optimal portfolios, particularly at low levels of risk aversion, effectively nullifying any theoretical advantage of direct utility maximization. Chopra and Ziemba (1993) show that the estimation errors in expected returns have ten times the impact of estimation errors in variances, and twenty times the impact of estimation errors in covariances.

[15] The expected return of an index is the market–value weighted average of the expected return of its constituents, and its volatility is given by a quadratic form that uses as inputs the market–value weights and the covariance matrix of its constituents.

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